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FLOATING ACTUATOR FOR MOWER DECK

Abstract

A mower includes a chassis, a tractive element coupled to the chassis, a mower deck coupled to the chassis and including a cutting element, and a deck actuator assembly configured to raise the mower deck relative to the chassis. The deck actuator assembly includes a frame coupled to the chassis, an output interface coupled to the mower deck and slidably coupled to the frame, a first stop fixedly coupled to the output interface, a second stop, and an actuator configured to reposition the second stop relative to the frame. The second stop is positioned to (a) engage the first stop to limit movement of the output interface in a first direction and (b) permit movement of the first stop in a second direction opposite the first direction.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates generally to mowers. More specifically, the present disclosure relates to an actuator for a deck of a mower.

[0002] Mowers are used to maintain vegetation (e.g., grass, clover, weeds, etc.) at a desired height. To accomplish this, mowers include at least one mower deck having a cutting element that is driven by a motor. A cutting height of the mower deck may be set by an operator to provide a desired trimmed height of the vegetation. When traveling at high speeds (e.g., between jobsites), a user may raise the mower deck to avoid contact between the mower deck and the ground. The height of the mower deck may be set by a mower deck actuator.

SUMMARY

[0003] One embodiment relates to a mower. The mower includes a chassis, a tractive element coupled to the chassis, a mower deck coupled to the chassis and including a cutting element, and a deck actuator assembly configured to raise the mower deck relative to the chassis. The deck actuator assembly includes a frame coupled to the chassis, an output interface coupled to the mower deck and slidably coupled to the frame, a first stop fixedly coupled to the output interface, a second stop, and an actuator configured to reposition the second stop relative to the frame. The second stop is positioned to (a) engage the first stop to limit movement of the output interface in a first direction and (b) permit movement of the first stop in a second direction opposite the first direction.

[0004] Another embodiment relates to a mower. The mower includes a chassis, a mower deck coupled to the chassis and including a cutting element, and a deck actuator assembly configured to raise the mower deck relative to the chassis. The deck actuator assembly includes a frame coupled to the chassis, an actuator shaft coupled to the mower deck and slidably coupled to the frame, a shaft stop fixedly coupled to the actuator shaft, a control stop positioned to engage the shaft stop, and an actuator. The actuator shaft is configured to move freely relative to the frame throughout a longitudinal range of motion. An end of the longitudinal range of motion is defined by the shaft stop engaging the control stop. The actuator is configured to vary the longitudinal range of motion of the actuator shaft by repositioning the control stop relative to the frame.

[0005] Still another embodiment relates to a floating linear actuator. The floating linear actuator includes a frame defining a first connection interface, a guide defining a shaft passage extending longitudinally through the guide, a standoff extending longitudinally between the frame and the guide and fixedly coupling the guide to the frame, an actuator shaft defining a second connection interface and extending through the shaft passage, a shaft stop fixedly coupled to the actuator shaft, a control stop defining a threaded passage and positioned to engage the shaft stop to limit movement of the shaft stop in a first direction and permit movement of the shaft stop in a second direction, a threaded rod rotatably coupled to the frame and extending through the threaded passage, and a motor configured to drive rotation of the threaded rod.

[0006] This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a perspective view of a mower, according to an exemplary embodiment.

[0008] FIG. 2 is a schematic block diagram of the mower of FIG. 1, according to an exemplary embodiment.

[0009] FIG. 3 is a side view of a connection between a mower deck and a frame of the mower of FIG. 1, according to an exemplary embodiment.

[0010] FIGS. 4, 5, and 6 are side views of a mower deck actuator of the mower of FIG. 1 at various stages of operation, according to an exemplary embodiment.

[0011] FIGS. 7, 8, and 9 are perspective views of the mower deck actuator of FIG. 4 at various stages of operation, according to an exemplary embodiment.

[0012] FIG. 10 is a perspective view of a transmission of the mower deck actuator of FIG. 7.

[0013] FIG. 11 is a schematic diagram of the transmission of FIG. 10.

[0014] FIGS. 12 and 13 are perspective views of sensors of the mower deck actuator of FIG. 7.

[0015] FIGS. 14 and 15 are perspective views of the mower deck actuator of FIG. 4 at various stages of operation, according to another exemplary embodiment.

[0016] FIG. 16 is a perspective view of a transmission of the mower deck actuator of FIG. 14.

[0017] FIG. 17 is a perspective view of the mower deck actuator of FIG. 4, according to another exemplary embodiment.

[0018] FIG. 18 is a top view of the mower deck actuator of FIG. 17.

[0019] FIG. 19 is another perspective view of the mower deck actuator of FIG. 17.

[0020] FIGS. 20 and 21 are front views of the mower deck actuator of FIG. 17 at various stages of operation.

[0021] FIGS. 22 and 23 are top view of the mower deck actuator of FIG. 17 at various stages of operation.

[0022] FIGS. 24, 25, 26, and 27 are front views of the mower deck actuator of FIG. 17 at various stages of operation.

DETAILED DESCRIPTION

[0023] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0024] According to an exemplary embodiment, a mower of the present disclosure includes a chassis, a mower deck including a cutting element, and a deck actuator that controls a height of the mower deck relative to the chassis. During normal operation, the mower deck uses the cutting element to trim vegetation (e.g., grass, clover, weeds, etc.). A user may use the deck actuator to set a cutting height of the mower deck or to raise the mower deck to a travel position (e.g., a height where the mower deck is out of contact with the vegetation). When the mower deck is at a cutting height and the mower passes over an obstacle (e.g., an incline or decline in the ground surface, and object, etc.), the mower deck may be forced upward. The deck actuator is a floating actuator that prevents downward movement of the mower deck below the set height but permits free upward movement of the mower deck. Accordingly, the deck actuator permits the mower deck to quickly rise over the obstacle and return back to the set height without damage to any components or a gap in the cutting operation. Other mowers use deck actuators lacking this floating capability. Upon encountering an obstacle, these actuators resist the upward movement of the mower deck, producing stresses within the deck actuator and potentially causing a delay in the cutting operation.

Overall Vehicle

[0025] As shown in FIGS. 1 and 2, a machine or vehicle, shown as mower 10, includes a chassis, shown as frame 12; a body assembly, shown as body 20, coupled to the frame 12 and having an occupant portion or section, shown as occupant seating area 30; operator input and output devices, shown as operator controls 40, that are disposed within the occupant seating area 30; a drivetrain, shown as driveline 50, coupled to the frame 12 and at least partially disposed under the body 20; a

vehicle suspension system, shown as suspension system **60**, coupled to the frame **12** and one or more components of the driveline **50**; a vehicle braking system, shown as braking system **70**, coupled to one or more components of the driveline **50** to facilitate selectively braking the one or more components of the driveline **50**; and a vehicle control system, shown as control system **100**, coupled to the operator controls **40**, the driveline **50**, the suspension system **60**, the braking system **70**, and/or one or more external devices or systems positioned remote from the mower **10**, shown as remote system(s) **200**. The mower **10** further includes a series of implements or mower assemblies, shown as mower decks **80**. In other embodiments, the mower **10** includes more or fewer components.

[0026] According to an exemplary embodiment, the mower **10** is an off-road machine or vehicle. In some embodiments, the mower **10** is a lawnmower, a turf mower, a push mower, a ride-on mower, a stand-on mower, or another type of mower. In other embodiments, the off-road machine or vehicle is a lightweight or recreational machine or vehicle such as a golf cart, an all-terrain vehicle (“ATV”), a utility task vehicle (“UTV”), and/or another type of lightweight or recreational machine or vehicle. In some embodiments, the off-road machine or vehicle is a chore product such as aerator, turf sprayer, bunker rake, and/or another type of chore product (e.g., that may be used on a golf course).

[0027] According to the exemplary embodiment shown in FIG. **1**, the occupant seating area **30** includes a single seat, shown as driver seat **32**. In some embodiments, the occupant seating area **30** includes additional seats (e.g., a passenger seat, an additional row of seats, etc.). According to the exemplary embodiment shown in FIG. **1**, the driver seat **32** is laterally centered on the body **20** and facing forward. In some embodiments, the driver seat **32** is facing rearward or otherwise positioned. In some embodiments, the occupant seating area **30** is omitted (e.g., the mower **10** is configured as a push mower).

[0028] According to an exemplary embodiment, the operator controls **40** are configured to provide an operator with the ability to control one or more functions of and/or provide commands to the mower **10** and the components thereof (e.g., turn on, turn off, drive, turn, brake, engage various operating modes, raise/lower a mower deck **80**, etc.). As shown in FIGS. **1** and **2**, the operator controls **40** include a steering interface (e.g., a steering wheel, joystick(s), etc.), shown as steering wheel **42**, an accelerator interface and/or braking interface (e.g., a pedal, a throttle, etc.), shown as traction pedal **44**, and one or more additional interfaces, shown as operator interface **48**. The traction pedal **44** may be used to control the speed and direction of travel of the mower **10**. By way of example, pressing the traction pedal **44** in a first direction may cause the driveline **50** to move the mower **10** forward, and pressing the traction pedal **44** in an opposing direction may cause the driveline **50** to move the mower **10** rearward. Returning the traction pedal **44** to a middle or neutral position may cause the braking system **70** and/or the driveline **50** to slow or stop the mower **10** or to hold the mower **10** in place. Alternatively, the operator interface **48** may include a pair of handles that control the driveline **50** in a zero-turn configuration (e.g., a left joystick to control the left side of the driveline **50** and a right joystick to control a right side of the driveline **50**). The operator interface **48** may be used to control operation of the mower decks **80** (e.g., changing a cutting speed of a mower deck **80**, changing a cutting height of a mower deck **80**, etc.). The operator interface **48** may include one or more displays and one or more input devices. The one or more displays may be or include a touchscreen, an LCD display, a LED display, a speedometer, gauges, warning lights, etc. The one or more input device may be or include buttons, switches, knobs, levers, dials, etc.

[0029] According to an exemplary embodiment, the driveline **50** is configured to propel the mower **10**. As shown in FIGS. **1** and **2**, the driveline **50** includes a primary driver, shown as prime mover **52**, an energy storage device, shown as energy storage **54**, a first tractive assembly (e.g., axles, wheels, tracks, differentials, etc.), shown as rear tractive assembly **56**, and a second tractive assembly (e.g., axles, wheels, tracks, differentials, etc.), shown as front tractive assembly **58**. In

some embodiments, the driveline **50** is a conventional driveline whereby the prime mover **52** is an internal combustion engine and the energy storage **54** is a fuel tank. The internal combustion engine may be a spark-ignition internal combustion engine or a compression-ignition internal combustion engine that may use any suitable fuel type (e.g., diesel, ethanol, gasoline, natural gas, propane, etc.). In some embodiments, the driveline **50** is an electric driveline whereby the prime mover **52** is one or more electric motors and the energy storage **54** is a battery system. In some embodiments, the driveline **50** is a fuel cell electric driveline whereby the prime mover **52** is one or more electric motors and the energy storage **54** is a fuel cell (e.g., that stores hydrogen, that produces electricity from the hydrogen, etc.). In some embodiments, the driveline **50** is a hybrid driveline whereby (i) the prime mover **52** includes an internal combustion engine and an electric motor/generator and (ii) the energy storage **54** includes a fuel tank and/or a battery system.

According to the exemplary embodiment shown in FIG. **1**, the rear tractive assembly **56** includes rear tractive elements and the front tractive assembly **58** includes front tractive elements that are configured as wheels. In some embodiments, the rear tractive elements and/or the front tractive elements are configured as tracks. In some embodiments, the driveline **50** is omitted, and the mower **10** is propelled by an operator (e.g., the mower **10** is configured as a push mower).

[0030] According to an exemplary embodiment, the prime mover **52** is configured to provide power to drive the rear tractive assembly **56** and/or the front tractive assembly **58** (e.g., to provide front-wheel drive, rear-wheel drive, four-wheel drive, and/or all-wheel drive operations). In some embodiments, the driveline **50** includes a transmission device (e.g., a gearbox, a continuous variable transmission (“CVT”), etc.) positioned between (a) the prime mover **52** and (b) the rear tractive assembly **56** and/or the front tractive assembly **58**. The rear tractive assembly **56** and/or the front tractive assembly **58** may include a drive shaft, a differential, and/or an axle. In some embodiments, the rear tractive assembly **56** and/or the front tractive assembly **58** include two axles or a tandem axle arrangement. In some embodiments, the rear tractive assembly **56** and/or the front tractive assembly **58** are steerable (e.g., using the steering wheel **42**). In some embodiments, both the rear tractive assembly **56** and the front tractive assembly **58** are fixed and not steerable (e.g., employ skid steer operations). By way of example, the driveline **50** may include a hydrostatic transmission that permits independent driving of the left and right sides of the driveline **50**.

[0031] In some embodiments, the driveline **50** includes a plurality of prime movers **52**. By way of example, the driveline **50** may include a first prime mover **52** that drives the rear tractive assembly **56** and a second prime mover **52** that drives the front tractive assembly **58**. By way of another example, the driveline **50** may include a first prime mover **52** that drives a first one of the front tractive elements, a second prime mover **52** that drives a second one of the front tractive elements, a third prime mover **52** that drives a first one of the rear tractive elements, and/or a fourth prime mover **52** that drives a second one of the rear tractive elements. By way of still another example, the driveline **50** may include a first prime mover **52** that drives the front tractive assembly **58**, a second prime mover **52** that drives a first one of the rear tractive elements, and a third prime mover **52** that drives a second one of the rear tractive elements. By way of yet another example, the driveline **50** may include a first prime mover **52** that drives the rear tractive assembly **56**, a second prime mover **52** that drives a first one of the front tractive elements, and a third prime mover **52** that drives a second one of the front tractive elements.

[0032] According to an exemplary embodiment, the suspension system **60** includes one or more suspension components (e.g., shocks, dampers, springs, etc.) positioned between the frame **12** and one or more components (e.g., tractive elements, axles, etc.) of the rear tractive assembly **56** and/or the front tractive assembly **58**. In some embodiments, the mower **10** does not include the suspension system **60**.

[0033] According to an exemplary embodiment, the braking system **70** includes one or more braking components (e.g., disc brakes, drum brakes, in-board brakes, axle brakes, etc.) positioned to facilitate selectively braking one or more components of the driveline **50**. In some embodiments,

the one or more braking components include (i) one or more front braking components positioned to facilitate braking one or more components of the front tractive assembly **58** (e.g., the front axle, the front tractive elements, etc.) and (ii) one or more rear braking components positioned to facilitate braking one or more components of the rear tractive assembly **56** (e.g., the rear axle, the rear tractive elements, etc.). In some embodiments, the one or more braking components include only the one or more front braking components. In some embodiments, the one or more braking components include only the one or more rear braking components. In some embodiments, the one or more front braking components include two front braking components, one positioned to facilitate braking each of the front tractive elements. In some embodiments, the one or more rear braking components include two rear braking components, one positioned to facilitate braking each of the rear tractive elements. In some embodiments, the driveline **50** is a hydrostatic transmission that performs braking by using hydraulic motors to oppose movement of the tractive elements.

[0034] As shown in FIG. 2, the control system **100** includes a controller **110**, one or more sensors **120**, and a communications interface **130** (e.g., located on the mower **10**). In some embodiments, the control system includes the remote system(s) **200**. In one embodiment, the controller **110** is configured to selectively engage, selectively disengage, control, or otherwise communicate with components of the mower **10**. According to an exemplary embodiment, the controller **110** is coupled to (e.g., communicably coupled to) components of the operator controls **40** (e.g., the steering wheel **42**, the traction pedal **44**, the operator interface **48**, etc.), components of the driveline **50** (e.g., the prime mover **52**), components of the braking system **70**, the sensors **120**, the communications interface **130**, and the remote system(s) **200**. By way of example, the controller **110** may send and receive signals (e.g., control signals, location signals, etc.) with the components of the operator controls **40**, the components of the driveline **50**, the components of the braking system **70**, the sensors **120**, the communications interface **130**, and/or the remote system(s) **200**.

[0035] The controller **110** may be implemented as a general-purpose processor, an application specific integrated circuit (“ASIC”), one or more field programmable gate arrays (“FPGAs”), a digital-signal-processor (“DSP”), circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. According to the exemplary embodiment shown in FIG. 2, the controller **110** includes a processing circuit **112** and a memory **114**. The processing circuit **112** may include an ASIC, one or more FPGAs, a DSP, circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. In some embodiments, the processing circuit **112** is configured to execute computer code stored in the memory **114** to facilitate the activities described herein. The memory **114** may be any volatile or non-volatile or non-transitory computer-readable storage medium capable of storing data or computer code relating to the activities described herein. According to an exemplary embodiment, the memory **114** includes computer code modules (e.g., executable code, object code, source code, script code, machine code, etc.) configured for execution by the processing circuit **112**. In some embodiments, the controller **110** may represent a collection of processing devices. In such cases, the processing circuit **112** represents the collective processors of the devices, and the memory **114** represents the collective storage devices of the devices.

[0036] The sensors **120** may include various sensors positioned about the mower **10** to acquire information or data regarding operation of the mower **10** and/or the location thereof. By way of example, the sensors **120** may include an accelerometer, a gyroscope, a compass, a position sensor (e.g., a GPS sensor, etc.), suspension sensor(s), wheel sensors, an audio sensor or microphone, and/or other sensors to facilitate acquiring information or data regarding operation of the mower **10** and/or the location thereof. According to an exemplary embodiment, one or more of the sensors **120** are configured to facilitate detecting and obtaining vehicle telemetry data including position of the mower **10**, whether the mower **10** is moving, travel direction of the mower **10**, slope of the

mower **10**, speed of the mower **10**, vibrations experienced by the mower **10**, sounds proximate the mower **10**, suspension travel of components of the suspension system **60**, and/or other vehicle telemetry data.

[0037] The communications interface **130** may be configured to facilitate wireless communications with the remote system(s) **200**. By way of example, the communications interface **130** may be configured to employ one or more types of wireless communications protocols including Bluetooth, Wi-Fi, radio, cellular, and/or other suitable wireless communications protocols.

[0038] The remote systems **200** may be or include an off-site server-based system that monitors various global positioning system (“GPS”) information and/or real-time kinematics (“RTK”) information (e.g., position/location, speed, direction of travel, geofence related information, etc.) and provides GPS data and/or RTK data based on the GPS information and/or RTK information to the controller **110** of the mower **10** through the communications interface **130**. The remote systems **200** may additionally or alternatively be or include an on-site system (e.g., in a club house of a golf course, on the golf course, etc.) that communicates with the mower **10** via the communications interface **130**. The on-site system may collect data from the mower **10** that may be used by the operators of the site (e.g., for advanced scheduling purposes, to identify persons braking course guidelines or rules, etc.). The on-site system may also function as an intermediary between the mower **10** and the off-site server-based system (e.g., if the communications interface **130** does not have long-range wireless communications capabilities). In some embodiments, the remote system(s) **200** include a processing circuit, a memory, and a communications interface similar to the control system **100**. In some embodiments (e.g., when the remote system(s) **200** are part of the control system **100**), the control system **100** includes a plurality of processing circuits **112** (e.g., a first processing circuit for the controller **110** of the vehicle **10** and a second processing circuit for the remote system(s) **200**) and a plurality of memories **114** (e.g., a first memory for the controller **110** of the vehicle **10** and a second memory for the remote system(s) **200**).

Mower Deck Actuators

[0039] Referring to FIG. **1**, the mower **10** includes a series of mower decks **80**. Each mower deck **80** includes a deck, housing, or enclosure, shown as housing **82**, and a cutting element **84** (e.g., a blade, a flail, a reel, etc.) movably coupled to the housing **82**. The housing **82** may open downward to expose the cutting element **84** to vegetation below the housing **82**. A motor or actuator (e.g., an electric motor, a hydraulic motor, etc.), shown as mower motor **86**, is coupled to the housing **82** and drives movement (e.g., rotation, oscillation, etc.) of the cutting element **84**. While driven by the mower motor **86**, the cutting element **84** crushes, mulches, removes, or otherwise trims vegetation beneath the housing **82**. Alternatively, the cutting element **84** may be driven by the prime mover **52** (e.g., through a power take off).

[0040] FIG. **3** illustrates a connection between a mower deck **80** and the frame **12**. The mower deck **80** of FIG. **3** may represent any of the mower decks **80** of FIG. **1**. A linkage or connection, shown as deck support **88**, extends between the housing **82** and the frame **12**. A first end portion of the deck support **88** is pivotally coupled to the housing **82**, and a second end portion of the deck support **88** is pivotally coupled to the frame **12**. A mower deck actuator (e.g., a floating actuator or linear actuator), shown as deck actuator **300**, extends between the frame **12** and the deck support **88**. A first portion or upper portion of the deck actuator **300**, shown as control section **302**, defines a connection interface or first end portion **304** of the deck actuator **300** that is pivotally coupled to the frame **12**. A second portion or lower portion of the deck actuator **300**, shown as floating section **306**, defines a connection interface or second end portion **308** of the deck actuator **300** that is pivotally coupled to the deck support **88**. The deck actuator **300** extends longitudinally from the first end portion **304** to the second end portion **308**. Accordingly, the mower deck **80** is indirectly coupled to the deck actuator **300** and the frame **12** through the deck support **88**.

[0041] The deck actuator **300** is operatively coupled to (e.g., controlled by) the controller **110**. The deck actuator **300** permits control over a length of the deck actuator **300** (e.g., a distance between

the first end portion **304** and the second end portion **308**). By varying a length of the deck actuator **300**, the control system **100** controls a height of the mower deck **80** relative to the frame **12**. The deck actuator **300** may set a cutting height of the mower deck **80**. The cutting height represents a final height of vegetation that is trimmed by the mower deck **80**. The deck actuator **300** may move the mower deck **80** to a travel position above the cutting height, in which the mower deck **80** is moved out of engagement with the vegetation and the ground surface. The travel position may be used when the mower **10** is traveling between job sites and/or the user does not wish to be trimming vegetation.

[0042] The floating section **306** can be retracted within the control section **302** to reduce an overall length of the deck actuator **300** and raise the deck support **88**. The floating section **306** can be extended out of the control section **302** to increase an overall length of the deck actuator **300** and lower the deck support **88**. As the deck support **88** rotates upward or downward, the pivotal coupling of the housing **82** to the deck support **88** permits the force of gravity to maintain the mower deck **80** in a horizontal orientation.

[0043] Referring to FIGS. **4-6**, the deck actuator **300** is shown according to an exemplary embodiment. The control section **302** includes a first portion (e.g., a chassis, a base, a frame, one or more plates, etc.), shown as frame **310**. The frame **310** defines the first end portion **304** of the control section **302** and supports the other components of the control section **302**. A support or linear guide, shown as guide plate **312**, is longitudinally offset from the frame **310**. A connecting member, shown as standoff **314**, extends from the frame **310** to the guide plate **312** and fixedly couples the guide plate **312** to the frame **310**. An aperture or passage, shown as shaft passage **316**, extends longitudinally through the guide plate **312**.

[0044] The floating section **306** includes a sliding portion or rod, shown as actuator shaft **320**, that is slidably coupled to the guide plate **312**. As shown, the actuator shaft **320** extends through the shaft passage **316** defined by the guide plate **312**, such that the actuator shaft **320** and the guide plate **312** together form a linear bearing or linear guide. The shaft passage **316** may be formed by a bushing to permit free movement of the actuator shaft **320** through the shaft passage **316**. The actuator shaft **320** defines the second end portion **308** of the deck actuator **300**. By way of example, a passage may extend laterally through the actuator shaft **320** to receive a pin that pivotally couples the actuator shaft **320** to the deck support **88**. The actuator shaft **320** is shaped to form a stop, block, bumper, collar, or limiter, shown as shaft stop **322**, that is fixed to the portion of the actuator shaft **320** opposite the second end portion **308**. The shaft stop **322** extends radially and laterally outward beyond the actuator shaft **320**.

[0045] The control section **302** further includes a stop, block, bumper, collar, or limiter, shown as control stop **330**, positioned between the shaft stop **322** and the guide plate **312**. The actuator shaft **320** extends through a passage defined by the shaft stop **322**, slidably coupling the actuator shaft **320** to the control stop **330**. A linear actuator (e.g., an electric linear actuator, a hydraulic cylinder, a pneumatic cylinder, etc.), shown as actuator **332**, is coupled to the frame **310** and the control stop **330**. The actuator **332** may be operatively coupled to and controlled by the controller **110**. The actuator **332** may control the position of the control stop **330** relative to the frame **310** by moving the control stop **330** longitudinally relative to the frame **310**. The actuator **332** may selectively limit (e.g., prevent) movement of the control stop **330** relative to the frame. The control stop **330** defines a passage, shown as shaft passage **334**, that extends longitudinally therethrough. The shaft passage **334** receives the actuator shaft **320**, slidably coupling the actuator shaft **320** to the control stop **330**.

[0046] The deck actuator **300** includes a pair of sensors, shown as retraction sensor **340** and extension sensor **342**. The retraction sensor **340** and the extension sensor **342** are examples of the sensors **120**. As shown in FIG. **2**, the retraction sensor **340** and the extension sensor **342** operatively coupled to the controller **110**. The retraction sensor **340** is coupled to the frame **310**. The retraction sensor **340** may indicate a position of the shaft stop **322** relative to the frame **310**. By way of example, the retraction sensor **340** may be or include a limit switch that engages the shaft stop **322**

when the shaft stop **322** is in a fully retracted or fully raised position. The extension sensor **342** is coupled to the guide plate **312**. The extension sensor **342** may indicate a position of the control stop **330** relative to the guide plate **312**. By way of example, the extension sensor **342** may be or include a limit switch that engages the control stop **330** when the control stop **330** is in a fully extended or fully lowered position. In some embodiments, the retraction sensor **340** and/or the extension sensor **342** continuously indicate the relative positions of the shaft stop **322** and/or the control stop **330**. By way of example, the retraction sensor **340** may include a linear potentiometer coupled to the frame **310** and the shaft stop **322**. In some embodiments, the retraction sensor **340** and/or the extension sensor **342** are omitted.

[0047] Referring to FIGS. **4-6**, the deck actuator **300** is shown at various stages of operation. In operation, the actuator **332** varies the longitudinal position of the control stop **330**. In FIG. **4**, the control stop **330** is shown in a fully extended configuration (i.e., the control stop **330** is at an extended end of its range of motion). In the fully extended configuration, the control stop **330** engages the extension sensor **342**, and the extension sensor **342** may provide a signal to the controller **110** indicating the position of the control stop **330**. In response to receiving this signal, the controller **110** may stop the actuator **332** from moving the control stop **330** further in the extension direction. As shown in FIG. **4**, the shaft stop **322** engages the control stop **330**, preventing further movement of the actuator shaft **320**. Accordingly, the configuration illustrated in FIG. **4** may represent a fully extended configuration of the deck actuator **300** (i.e., where the deck actuator **300** is extended to the maximum allowable length).

[0048] In FIG. **5**, the deck actuator **300** is shown in a floating configuration where the shaft stop **322** is spaced away from the control stop **330**. In FIG. **5**, the control stop **330** remains in the fully extended configuration. The control stop **330** limits the range of motion of the actuator shaft **320** in the extension direction, but the actuator shaft **320** is free to move away from the control stop **330** (e.g., until the shaft stop **322** contacts the frame **310**). Once the shaft stop **322** is spaced away from the control stop **330** (e.g., as shown in FIG. **5**), the actuator shaft **320** can freely extend until the shaft stop **322** engages the control stop **330**. Accordingly, the actuator shaft **320** is free to move throughout a range of motion, but the range of motion is limited in an extension direction by the control stop **330**.

[0049] In FIG. **6**, the actuator **332** retracts, bringing the control stop **330** toward the frame **310** and away from the guide plate **312**. This movement of the control stop **330** reduces the range of motion of the actuator shaft **320**. If the shaft stop **322** initially starts beyond this reduced range of motion, the control stop **330** contacts the shaft stop **322** and forces the actuator shaft **320** to retract with the control stop **330**. In the configuration of FIG. **6**, the actuator shaft **320** is free to float throughout the reduced range of motion, but the range of motion does not extend as far in the extension direction as in the configuration of FIGS. **4** and **5**.

[0050] If the shaft stop **322** reaches a fully retracted position (e.g., in which the shaft stop **322** contacts or is close to contacting the frame **310**), the shaft stop **322** engages the retraction sensor **340**, and the retraction sensor **340** may provide a signal to the controller **110** indicating the position of the shaft stop **322**. If this signal is received while the actuator **332** is stationary or extending, the controller **110** may interpret the signal as an indication that the actuator shaft **320** is being retracted by an outside force and may take no action. If the signal is received while the actuator **332** is retracting the control stop **330** toward the frame **310**, the controller **110** may interpret the signal as an indication that the control stop **330** is in a fully retracted configuration. In the fully retracted configuration, the control stop **330** forces the shaft stop **322** into the fully retracted position, and the actuator shaft **320** may be fixed (i.e., may no longer be permitted to move throughout a range of motion). In response to receiving this signal, the controller **110** may stop the actuator **332** from moving the control stop **330** further in the retraction direction.

[0051] Alternatively, the retraction sensor **340** may be positioned to be contacted by the control stop **330** instead of the shaft stop **322**. In such a configuration, the retraction sensor **340** would

provide direct feedback regarding the position of the control stop **330**, regardless of outside forces on the actuator shaft **320**. For example, the shaft stop **322** could move upward due to the application of an outside force onto the actuator shaft **320** (e.g., an upward force on the mower deck **80**). If the retraction sensor **340** were positioned to be contacted by the control stop **330**, such an input would not be detected by the retraction sensor **340**, and the control stop **330** could be moved freely without interference by the controller **110**. If the retraction sensor **340** were instead positioned to be contacted by the shaft stop **322**, the controller **110** may limit movement of the control stop **330** in the retraction direction, even if the control stop **330** was not in the fully retracted configuration.

[0052] Referring to FIGS. **3-6**, the deck actuator **300** may be used to permit floating actuation of the mower deck **80**. Throughout operation of the mower **10**, it may be desirable to set a mower deck **80** at a user-specified height. By way of example, a user may set the mower deck **80** to an operating height (e.g., an operating position) when using the mower **10** to trim vegetation. By way of another example, the user may raise the mower deck **80** from the operating height to a travel height (e.g., a travel position) above the cutting height. By raising the mower deck **80**, the mower deck **80** is moved above a ground surface, preventing inadvertent trimming operations and permitting travel without the mower deck **80** contacting obstacles.

[0053] During a mowing operation, the mower deck **80** may be lowered to the operating height, and the mower motor **86** may be activated. A cutting height between the mower deck **80** and the ground surface may be predefined by the user based on the desired height of the vegetation trimmed by the mower **10**. In some embodiments, the housing **82** includes a set of wheels that engage the ground surface to set the cutting height. In some embodiments, the position of the control stop **330** is set to define the cutting height.

[0054] At various points throughout operation, the mower deck **80** may encounter an obstacle (e.g., an incline or decline in the ground surface, a stone, a branch, etc.). As the mower **10** drives over the obstacle, the obstacle may force the mower deck **80** upward. In response to such an upward force, the actuator shaft **320** is forced in a retracting direction relative to the frame **310**. The shaft stop **322** moves freely away from the control stop **330**, such that the deck actuator **300** freely retracts to accommodate the upward force of the obstacle. When the mower **10** has passed the obstacle, the shaft stop **322** permits the actuator shaft **320** to freely extend due to the force of gravity on the mower deck **80** until the shaft stop **322** engages the control stop **330**. In this way, the deck actuator **300** executes a floating control over the position of the mower deck **80** in which the deck actuator **300** permits free upward movement of the mower deck **80** and prevents downward movement of the mower deck **80** beyond the position set by the control stop **330**. Because the actuator shaft **320** is free floating, the deck actuator **300** may not limit an upward or downward speed of the mower deck **80**.

[0055] Other mower deck actuators do not have the floating capabilities of the deck actuator **300**. Instead, the other actuators actively control the vertical position of a mower deck at all times, resisting vertical movement of the mower deck driven by an external force (e.g., from the obstacle). These actuators may limit the upward speed of the mower deck, such that the actuator undergoes significant forces when encountering an obstacle (e.g., potentially damaging the actuator, potentially causing the electric motor of the actuator to generate undesired electrical energy, etc.). These actuators may also limit the downward speed of the mower deck, which prevents the mower deck from immediately returning to the operating height after passing an obstacle and causes a disruption in the cutting operation.

[0056] In FIGS. **4 6**, the control stop **330** is shown to limit extension of the deck actuator **300** and permit free retraction of the deck actuator **300**. In other embodiments, the deck actuator **300** is configured to limit retraction of the deck actuator **300** and permit free extension of the deck actuator **300**. By way of example, the control stop **330** may be positioned between the frame **310** and the shaft stop **322**. Such a control stop **330** may be used as an alternative to the position of the

control stop **330** shown in FIGS. **4-6**. Alternatively, a first control stop **330** may be positioned between the frame **310** and the shaft stop **322**, and a second control stop **330** may be positioned between the guide plate **312** and the shaft stop **322**. In such an embodiment, each control stop **330** may be independently controlled by a separate actuator to independently vary a different end of the range of motion of the actuator shaft **320**.

[0057] One possible application for a deck actuator **300** that is configured to limit retraction and permit free extension is as a secondary braking system in a vehicle (e.g., a train or automobile). In such an example, the actuator shaft **320** may be coupled to a braking element, such as a brake caliper. During normal operation, the deck actuator **300** would have a large permitted range of motion, and the position of the actuator shaft **320** would be controlled by a primary braking system (e.g., a pneumatic or hydraulic braking system). When desired (e.g., in response to a demand for a parking brake or emergency brake), the actuator **332** would override the primary braking system by reducing the range of motion and forcing the actuator shaft **320** outward to engage the brake.

Deck Actuator With Threaded Rods

[0058] Referring to FIGS. **7-13**, a deck actuator **300** is shown according to an exemplary embodiment. The deck actuator **300** shown in FIGS. **7-13** represents one possible configuration of the deck actuator **300** of FIGS. **4-6**. Accordingly, any description of with respect to the deck actuator **300** shown in FIGS. **4-6** may apply to the deck actuator **300** of FIGS. **7-13**, except as otherwise specified herein.

[0059] The frame **310** of the deck actuator **300** includes a series of plates. Specifically, the frame **310** includes an end plate **400**, a middle plate **402**, and an end plate **404** extending parallel to one another. The middle plate **402** extends between the end plate **400** and the end plate **404**. A support plate **406** extends perpendicular to the end plate **400**, the middle plate **402**, and the end plate **404**. The end plate **400**, the middle plate **402**, and the end plate **404** are each fixedly coupled to the support plate **406** and to one another (e.g., through welding, fastening, etc.). A series of standoffs may extend between the end plate **400**, the middle plate **402**, and/or the end plate **404** to maintain a first spacing between the end plate **400** and the middle plate **402**, and a second spacing between the middle plate **402** and the end plate **404**.

[0060] The deck actuator **300** includes a pair of standoffs **314** that extend between the end plate **404** and the guide plate **312**. The standoffs **314** fixedly couple the end plate **404** to the guide plate **312** and maintain a fixed distance between the end plate **404** and the guide plate **312**. By way of example, the standoffs **314** may each be in threaded engagement with the guide plate **312** and the end plate **404**. The standoffs **314** may each be formed as cylindrical rods.

[0061] The control stop **330** forms a first pair of apertures or passages, shown as threaded passages **410**. The threaded passages **410** each extend longitudinally through the control stop **330**. The threaded passages **410** are each internally threaded. By way of example, the threaded passages **410** may each be at least partially defined by a nut of the control stop **330**. The control stop **330** forms a second pair of apertures or passages, shown as through holes **412**, that each extend longitudinally through the control stop **330**. Each through hole **412** is sized to receive one of the standoffs **314**. The relative sizes and shapes of the through holes **412** and the standoffs **314** may be matched to permit free sliding movement of the control stop **330** along the standoffs **314**. By way of example, the standoffs **314** may have a circular cross section, and the through holes **412** may be circular and a clearance fit for the standoffs **314**. The threaded passage **410** and the through holes **412** are arranged in a square pattern centered about the actuator shaft **320**. Accordingly, the threaded passages **410** and the through holes **412** are each equidistant from the actuator shaft **320**. The threaded passages **410** are arranged on opposite sides of the actuator shaft **320**, and the through holes **412** are arranged on opposite sides of the actuator shaft **320**.

[0062] The shaft stop **322** forms four apertures or passages, shown as through holes **420**, that each extend longitudinally through the shaft stop **322**. Each through hole **420** is sized to receive one of the standoffs **314** or one of the threaded rods **442** described herein. Each through hole **420** is

aligned with one of the through holes **412** or one of the threaded passages **410**.

[0063] The actuator **332** includes a motor (e.g., an electric motor, a hydraulic motor, etc.) shown as electric motor **440**, and a pair of threaded rods or screws, shown as threaded rods **442**. The electric motor **440** drives rotation of the threaded rods **442** to control the position of the control stop **330**. The threaded rods **442** extend from the end plate **404** to the guide plate **312** and are rotatably coupled to the end plate **404** and the guide plate **312**. Each threaded rod **442** extends through one of the through holes **420** and one of the threaded passages **410**.

[0064] The electric motor **440** is rotationally coupled to the threaded rods **442** by a gearbox, power transmission, driveline, or gear train, shown as transmission **450**. The transmission **450** transfers rotational mechanical energy from the electric motor **440** to the threaded rods **442**. Accordingly, rotation of the electric motor **440** causes a corresponding rotation of both threaded rods **442**. Due to engagement between the threaded rods **442** and the threaded passages **410**, the rotation of the threaded rods **442** causes a corresponding linear movement of the control stop **330** along the standoffs **314**. Rotation of the threaded rods **442** in a first direction causes the control stop **330** to move in an extension direction. Rotation of the threaded rods **442** in an opposing, second direction causes the control stop **330** to move in a retraction direction. Accordingly, the controller **110** may control the position of the control stop **330** by controlling rotation of the electric motor **440** (e.g., by changing a voltage, current, frequency, and/or polarity of electrical energy supplied to the electric motor **440**).

[0065] Referring to FIGS. **10** and **11**, the transmission **450** is shown according to an exemplary embodiment. The transmission **450** has a two-stage arrangement formed using a series of spur gears. A first stage is positioned between the middle plate **402** and the end plate **404**. The first stage includes a pinion gear **452** in meshing engagement with a gear **454**. The gear **454** is fixedly coupled to a first one of the threaded rods **442**. The pinion gear **452** is coupled to an output shaft of the electric motor **440**. The pinion gear **452** is smaller in diameter than the gear **454**, reducing an output speed and increasing an output torque of the transmission **450**.

[0066] A second stage is positioned between the end plate **400** and the middle plate **402**. The second stage includes a screw gear **460** that is fixedly coupled to the gear **454** and the first one of the threaded rods **442**, such that the screw gear **460**, the gear **454**, and the first one of the threaded rods **442** rotate at the same speed and in the same direction. An idler gear **462** is in meshing engagement with the screw gear **460** and a screw gear **464**. The screw gear **464** is fixedly coupled to a second one of the threaded rods **442**. Accordingly, the transmission **450** transfers rotational mechanical energy from the pinion gear **452** to the screw gear **460** and the screw gear **464**.

[0067] Due to the position of the idler gear **462** between the screw gear **460** and the screw gear **464**, the idler gear **462** causes the screw gear **460** and the screw gear **464** to rotate in the same direction. Accordingly, both of the threaded rods **442** may be threaded in the same direction (e.g., both right hand thread, both left hand thread, etc.). In other embodiments, the idler gear **462** is omitted, and the screw gear **460** directly engages the screw gear **464**. In such an embodiment, the screw gear **460** and the screw gear **464** rotate in opposite directions. Accordingly, the threaded rods **442** may be threaded in opposing directions (e.g., one left hand thread and one right hand thread).

[0068] In some embodiments, the actuator **332** is configured to hold the control stop **330** in place under the load of the actuator shaft **320** without requiring a supply of energy from an outside source (e.g., without electrical energy being supplied to the electric motor **440**). It may be desirable for the deck actuator **300** to hold the mower deck **80** at the desired height until the desired height changes, even if the mower **10** is powered off. This may reduce wear on components (e.g., from repeatedly having to raise the mower deck **80** back to the desired position) and/or the burden on the operator (e.g., from having to repeatedly confirm the position of the mower deck **80**). In some embodiments, the pitch of the threads on the threaded rods **442** is selected such that the frictional forces between the threads of the control stop **330** and the threaded rods **442** is sufficient to prevent backdriving of the threaded rods **442** when applying a longitudinal force on the control stop **330**.

An exemplary configuration of the deck actuator **300** was tested in which the mating surfaces of the threaded rods **442** and the control stop **330** were both steel having a coefficient of static friction of 0.25. In this testing, it was determined that a thread pitch of 15 degrees or less was sufficient to hold the control stop **330** in place as desired.

[0069] FIGS. 7-9 illustrate the deck actuator **300** at various stages of operation. In FIG. 7, the shaft stop **322** engages the control stop **330**, preventing further extension of the actuator shaft **320**. In FIG. 8, the shaft stop **322** has moved freely away from the control stop **330** due to a compressive force on the deck actuator **300** (e.g., an upward force on the mower deck **80**). In FIG. 9, the electric motor **440** has driven the control stop **330** toward the frame **310**, limiting the range of motion of the actuator shaft **320**.

[0070] Referring to FIGS. 14-16, a deck actuator **300** is shown according to an exemplary embodiment. The deck actuator **300** shown in FIGS. 14-16 represents another possible configuration of the deck actuator **300** of FIGS. 4-6 and may be substantially similar to the deck actuator **300** of FIGS. 7-13. Any description of with respect to the deck actuator **300** shown in FIGS. 4-6 or the deck actuator **300** of FIGS. 7-13 may apply to the deck actuator **300** of FIGS. 14-16, except as otherwise specified herein.

[0071] The deck actuator **300** of FIGS. 14-16 includes two actuator shafts **320** each fixedly coupled to the shaft stop **322** and each extending through a shaft passage **334** defined by the control stop **330**. Although the standoffs **314** and the guide plate **312** are omitted from FIGS. 14-16, it should be understood that the deck actuator **300** may include a pair of standoffs **314** and a guide plate **312**, and the actuator shafts **320** may each be received through a shaft passage **316** of the guide plate **312**.

[0072] The deck actuator **300** of FIGS. 14-16 includes a transmission **500** in place of the transmission **450**. The transmission **500** is similar to the transmission **450**, except the transmission **500** has a single-stage arrangement. A pinion gear **502** is coupled to an output shaft of the electric motor **440**. The pinion gear **502** is in direct meshing engagement with a pair of screw gears **504**, and each screw gear **504** is rotationally fixed to one of the threaded rods **442**. The pinion gear **502** is smaller in diameter than the screw gears **504**, reducing an output speed and increasing an output torque of the transmission **500**. Because the pinion gear **502** directly engages the screw gears **504**, both threaded rods **442** rotate in the same direction.

Deck Actuator With Linkages

[0073] Referring to FIGS. 17-27, a deck actuator **300** is shown according to an exemplary embodiment. The deck actuator **300** shown in FIGS. 17-27 represents another possible configuration of the deck actuator **300** of FIGS. 4-6 and may be substantially similar to the deck actuator **300** of FIGS. 7-13. Any description of with respect to the deck actuator **300** shown in FIGS. 4-6 or the deck actuator **300** of FIGS. 7-13 may apply to the deck actuator **300** of FIGS. 17-27, except as otherwise specified herein.

[0074] The deck actuator **300** of FIGS. 17-27 includes one control section **302** and a pair of floating sections **306**. The control section **302** is positioned between the floating sections **306** and receives each floating section **306**. For ease of description, one of the floating sections **306** and portions of the control section **302** will be referred to as a first end assembly **600**, and the other of the floating sections **306** and portions of the control section **302** will be referred to as a second end assembly **602**. The floating section **306** of the first end assembly **600** defines the first end portion **304**, and the floating section **306** of the second end assembly **602** defines the second end portion **308**. Accordingly, the deck actuator **300** may be coupled to the frame **12** and the deck support **88** by the floating sections **306**.

[0075] Referring to FIGS. 17-19, the frame **310** of the deck actuator **300** includes a pair of guide plates **312** longitudinally offset from one another, each belonging to one of the end assemblies **600** and **602**. The guide plates **312** may be placed such that the shaft passages **316** are aligned with one another (e.g., are centered about a common longitudinal axis). A pair of support members, shown as

guide plate spacers **606**, extend longitudinally between the guide plates **312**. One guide plate spacer **606** is positioned at the top of the guide plates **312**, and the other guide plate spacer **606** is positioned at the bottom of the guide plates **312**. An end portion of each guide plate spacer **606** is fixedly coupled to one of the guide plates **312**. The guide plate spacers **606** facilitate maintaining a fixed distance and orientation between the guide plates **312**.

[0076] The deck actuator **300** includes four standoffs **314** that extend between the guide plates **312**. The standoffs **314** fixedly couple the guide plates **312** to one another and maintain a fixed distance between the guide plates **312**. By way of example, the standoffs **314** may each be in threaded engagement with the guide plates **312**. The standoffs **314** may each be formed as cylindrical rods. The standoffs **314** are arranged in a square pattern centered about the shaft passages **316**.

[0077] The frame **310** further includes a pair of plates or supports, shown as support plate **610** and support plate **612**. The support plates **610** and **612** are laterally offset from one another, such that the standoffs **314** and the guide plate spacers **606** extend between the support plates **610** and **612**. A pair of support members, shown as lateral supports **614**, extend laterally between the support plates **610** and **612**. The lateral supports **614** are each fixedly coupled to one of the guide plate spacers **606**. The guide plates **312**, the standoffs **314**, the guide plate spacers **606**, the support plates **610** and **612**, and the lateral supports **614** are all fixedly coupled to one another (e.g., with fasteners, by welding, etc.) to form the frame **310**.

[0078] The deck actuator **300** includes a pair of actuator shafts **320**, each belonging to one of the end assemblies **600** and **602**. Each actuator shaft **320** is positioned within one of the shaft passages **316**, slidably coupling the actuator shaft **320** to the corresponding guide plate **312**. A first actuator shaft **320** extends away from the guide plate **312** in a first direction, and a passage extends laterally through an end portion of the first actuator shaft **320** to define the first end portion **304**. A second actuator shaft **320** extends away from the guide plate **312** in a second direction opposite the first direction, and a passage extends laterally through an end portion of the second actuator shaft **320** to define the second end portion **308**. As shown in FIGS. **20** and **21**, the actuator shaft **320** of the first end assembly **600** is longer than the actuator shaft **320** of the second end assembly **602**. In other embodiments, the actuator shafts **320** are otherwise sized.

[0079] In some embodiments, the actuator shafts **320** are both centered about and movable along a longitudinal axis of extension, shown as axis **620**. The actuator shafts **320** may be movable independent of one another. By way of example, the actuator shafts **320** may move in the same direction or in opposing directions. By way of another example, one actuator shaft **320** may move relative to the frame **310** while the other actuator shaft **320** remains stationary. The configuration illustrated in FIG. **20** may represent a fully extended configuration of the deck actuator **300** (i.e., where the deck actuator **300** is extended to the maximum allowable length). The configuration illustrated in FIG. **21** may represent a retracted configuration of the deck actuator **300** (i.e., where the deck actuator **300** is retracted to less than the maximum allowable length).

[0080] A stop, block, bumper, collar, or limiter, shown as shaft stop **322**, is fixedly coupled to each actuator shaft **320**. The shaft stops **322** each extend radially and laterally outward beyond the corresponding actuator shaft **320**. A first shaft stop **322** is coupled to an end portion of the first actuator shaft **320** opposite the first end portion **304**. A second shaft stop **322** is coupled to an end portion of the second actuator shaft **320** opposite the second end portion **308**. The guide plate **312** of the first end assembly **600** is positioned between the first shaft stop **322** and the first end portion **304**. The guide plate **312** of the second end assembly **602** is positioned between the second shaft stop **322** and the second end portion **308**. Accordingly, the shaft stops **322** are both positioned between the guide plates **312**.

[0081] Each shaft stop **322** forms one or more apertures or passages, shown as through holes **420**, that each extend longitudinally through the shaft stop **322**. Each through hole **420** is sized to receive one of the standoffs **314**. The standoffs **314** extend through the through holes **420**, guiding movement of the actuator shafts **320** along the axis **620** and limiting rotation of the actuator shafts

320.

[0082] The control section **302** includes a pair of stops, blocks, bumpers, collars, or limiters, shown as control stops **330**, each belonging to one of the end assemblies **600** and **602**. Each control stop **330** defines a passage, shown as shaft passage **334**, that extends longitudinally therethrough. In the first end assembly **600**, a control stop **330** is positioned between a guide plate **312** and a shaft stop **322**, with the control stop **330** positioned farther in a first longitudinal direction than the shaft stop **322**. An actuator shaft **320** coupled to the shaft stop **322** extends through the shaft passage **334** of the control stop **330**, slidably coupling the control stop **330** to the actuator shaft **320**. Engagement between the shaft stop **322** and the control stop **330** limits movement of the actuator shaft **320** in the first longitudinal direction, while permitting free movement of the actuator shaft **320** in an opposing second longitudinal direction.

[0083] The second end assembly **602** has a similar arrangement, but oriented in an opposing direction. In the second end assembly **602**, a control stop **330** is positioned between a guide plate **312** and a shaft stop **322**, with the control stop **330** positioned farther in the second longitudinal direction than the shaft stop **322**. An actuator shaft **320** coupled to the shaft stop **322** extends through the shaft passage **334** of the control stop **330**, slidably coupling the control stop **330** to the actuator shaft **320**. Engagement between the shaft stop **322** and the control stop **330** limits movement of the actuator shaft **320** in the second longitudinal direction, while permitting free movement of the actuator shaft **320** in the first longitudinal direction.

[0084] Referring to FIGS. **19-23**, a rotational actuator assembly (e.g., an electric actuator assembly, a hydraulic motor assembly, a pneumatic motor assembly, etc.), shown as actuator **630**, is coupled to the frame **310** and the control stops **330**. The actuator **630** may be operatively coupled to and controlled by the controller **110**. Similar to the actuator **332**, the actuator **630** may control the position of the control stops **330** relative to the frame **310** by moving the control stops **330** longitudinally relative to the frame **310**. The actuator **630** may selectively limit (e.g., prevent) movement of the control stops **330** relative to the frame **310**.

[0085] The actuator **630** includes a rotational linkage assembly, shown as control linkage **640**, coupled to the control stops **330**. The control linkage **640** includes a pair of plates, discs, or rotating members, shown as drive wheel **642** and driven wheel **644**. The drive wheel **642** and the driven wheel **644** are laterally offset from one another and positioned between the support plate **610** and the support plate **612**. The drive wheel **642** and the driven wheel **644** are rotatably coupled to the frame **310** by a shaft, shown as support shaft **646**. The support shaft **646** extends laterally across the frame **310**. A first end portion of the support shaft **646** is rotatably coupled to the support plate **610** by a rotational support (e.g., a bearing, a bushing, etc.), shown as bearing **648**, and a second end portion of the support shaft **646** is rotatably coupled to the support plate **612** by the transmission **690**.

[0086] The drive wheel **642** and the driven wheel **644** may be rotatable relative to the frame **310** about an axis of rotation, shown as axis **650**, that extends laterally and is centered about the support shaft **646**. In some embodiments, the drive wheel **642** and the driven wheel **644** are fixedly coupled to the support shaft **646** (e.g., through welding), to transfer torque between the drive wheel **642** and the driven wheel **644** and maintain a common rotational orientation of the drive wheel **642** and the driven wheel **644** about the axis **650**. By way of example, if the drive wheel **642** rotates 20 degrees clockwise, the driven wheel **644** may also rotate 20 degrees clockwise.

[0087] The control linkage **640** includes a first pair of linkage members, shown as links **660**, that belong to the first end assembly **600**. The links **660** couple the control stop **330** of the first end assembly **600** to the drive wheel **642** and the driven wheel **644**. Specifically, a first link **660** is pivotally coupled to the control stop **330** and the drive wheel **642**, and a second link **660** is pivotally coupled to the control stop **330** and the driven wheel **644**. By way of example, each link **660** may define (a) a first passage that receives a pin to rotatably couple the link **660** to the control stop **330** and (b) a second passage that receives a pin to rotatably couple the link **660** to the drive

wheel **642** or the driven wheel **644**. The links **660** are pivotable relative to the control stop **330** about a first lateral axis of rotation, shown as axis **662**. The links **660** are pivotable relative to the drive wheel **642** and the driven wheel **644** about a second lateral axis of rotation, shown as axis **664**.

[0088] The control linkage **640** includes a second pair of linkage members, shown as links **670**, that belong to the second end assembly **602**. The links **670** couple the control stop **330** of the second end assembly **600** to the drive wheel **642** and the driven wheel **644**. Specifically, a first link **670** is pivotally coupled to the control stop **330** and the drive wheel **642**, and a second link **670** is pivotally coupled to the control stop **330** and the driven wheel **644**. By way of example, each link **670** may define (a) a first passage that receives a pin to rotatably couple the link **670** to the control stop **330** and (b) a second passage that receives a pin to rotatably couple the link **670** to the drive wheel **642** or the driven wheel **644**. The links **670** are pivotable relative to the control stop **330** about a first lateral axis of rotation, shown as axis **672**. The links **670** are pivotable relative to the drive wheel **642** and the driven wheel **644** about a second lateral axis of rotation, shown as axis **674**.

[0089] The actuator **630** includes a motor (e.g., an electric motor, a hydraulic motor, etc.), shown as electric motor **440**, that drives rotation of the drive wheel **642** to control the positions of the control stops **330**. The electric motor **440** is coupled to a gearbox, power transmission, driveline, or gear train, shown as transmission **690**. The transmission **690** may include one or more gears or other power transmission members that transfer rotational mechanical energy from the electric motor **440** to an output member, shown as pinion gear **692**. The pinion gear **692** is in engagement with the drive wheel **642**, linking rotation of the pinion gear **692** and the drive wheel **642**. By way of example, the pinion gear **692** may include external gear teeth in meshing engagement with corresponding external gear teeth on the drive wheel **642**. Accordingly, rotation of the electric motor **440** causes a corresponding rotation of the drive wheel **642**. The transmission **690** may include components (e.g., gears) that reduce an output speed of the pinion gear **692** and increase an output torque of the pinion gear **692** relative to that of the electric motor **440**. Similarly, the pinion gear **692** may be smaller in diameter than the drive wheel **642** to reduce the speed of the drive wheel **642** and increase a torque on the drive wheel **642**.

[0090] During operation, the electric motor **440** causes rotation of the drive wheel **642** and the driven wheel **644**. This rotation pulls or pushes the links **660** and the links **670**, which in turn cause movement of the control stops **330** along the axis **620**. Specifically, rotation of the drive wheel **642** and the driven wheel **644** clockwise as shown in FIGS. **20** and **21** causes the control stops **330** to move toward one another. Rotation of the drive wheel **642** and the driven wheel **644** counterclockwise as shown in FIGS. **20** and **21** causes the control stops **330** to move away from one another. Accordingly, the controller **110** may control the positions of the control stops **330** by controlling rotation of the electric motor **440** (e.g., by changing a voltage, current, frequency, and/or polarity of electrical energy supplied to the electric motor **440**).

[0091] As shown in FIGS. **20** and **21**, the control linkage **640** includes a protrusion or registration member, shown as sensor bumper **700**, that is fixedly coupled to the driven wheel **644**. As shown, the sensor bumper **700** is formed as a single continuous piece with the driven wheel **644**. In other embodiments, the sensor bumper **700** is otherwise coupled to the driven wheel **644** (e.g., indirectly through the drive wheel **642** and the support shaft **646**). The sensor bumper **700** is radially offset from the axis **650**, such that the sensor bumper **700** is movable between (a) a first position shown in FIG. **20** and (b) a second position shown in FIG. **21**. Due to the connections of the links **660** and **670**, the position of the sensor bumper **700** corresponds to the positions of the control stops **330**. Specifically, the position shown in FIG. **20** corresponds to the fully extended positions of the control stops **330**, and the position shown FIG. **21** corresponds to retracted positions or travel positions of the control stops **330**.

[0092] The deck actuator **300** includes a retraction sensor **340** and an extension sensor **342**, each

coupled to the frame **310**. The retraction sensor **340** and the extension sensor **342** are positioned to contact the sensor bumper **700**. Specifically, the retraction sensor **340** is positioned to contact the sensor bumper **700** when the control stops **330** are in the travel positions. The extension sensor **342** is positioned to contact the sensor bumper **700** when the control stops **330** are in the fully extended positions. In other embodiments, the deck actuator **300** includes a different type of sensor. By way of example, the retraction sensor **340** may include a rotary potentiometer coupled to the frame **310** and the driven wheel **644**.

[0093] Referring to FIGS. **24-27**, the deck actuator **300** is shown at various stages of operation. The driven wheel **644** is omitted from FIGS. **24-27** for ease of illustration. In operation, the actuator **332** varies the positions of the control stops **330** along the axis **620**. In FIG. **24**, the control stops **330** are shown in a fully extended configuration (i.e., each control stop **330** is at an extended end of its range of motion). In the fully extended configuration, the sensor bumper **700** engages the extension sensor **342** (as shown in FIG. **20**), and the extension sensor **342** may provide a signal to the controller **110** indicating the positions of the control stops **330**. In response to receiving this signal, the controller **110** may stop the actuator **630** from moving the control stops **330** further in the extension directions (e.g., by preventing further counterclockwise rotation of the drive wheel **642**). As shown in FIG. **24**, the shaft stops **322** are offset from the respective control stops, permitting movement of each actuator shaft **320** in both a retraction direction and an extension direction. Accordingly, FIG. **24** may represent a floating configuration of the deck actuator **300**.

[0094] In FIG. **25**, the drive wheel **642** is rotated clockwise, retracting both of the control stops **330** inward. The shaft stops **322** each engage one of the control stops **330**, limiting outward movement of the actuator shafts **320**. In order to counteract an outward or extending force on the actuator shafts **320** (e.g., caused by gravity acting on a mower deck **80**), the links **660** and **670** apply an opposing force on the control stops **330**, placing the links **660** and **670** in tension. The links **660** impart a force F on the drive wheel **642**, the force F acting along a direction intersecting the axis **662** and the axis **664**. The links **670** impart a force F on the drive wheel **642**, the force F acting along a line intersecting the axis **672** and the axis **674**. Due to the positions of the links **660** and **670** relative to the axis **650**, these forces F impart a resultant moment or torque, shown as moment M , acting in a positive direction (i.e., counterclockwise) on the drive wheel **642**. Accordingly, the electric motor **440** may apply a torque or moment in a negative direction to hold the positions of the control stops **330** or further retract the control stops **330**.

[0095] In FIG. **26**, the drive wheel **642** is rotated further clockwise, retracting both of the control stops **330** further inward. The shaft stops **322** each engage one of the control stops **330**, limiting outward movement of the actuator shafts **320**. The links **660** and **670** continue to apply a retracting force on the control stops **330**, placing the links **660** and **670** in tension. The links **660** impart a force F on the drive wheel **642**, and the links **670** impart a force F on the drive wheel **642**. Due to the positions of the links **660** and **670** relative to the axis **650**, the moment M resultant from the forces F continues to act in a positive direction (i.e., counterclockwise) on the drive wheel **642**. Accordingly, the electric motor **440** applies a torque or moment in a negative direction to hold the positions of the control stops **330** or further retract the control stops **330**.

[0096] As the drive wheel **642** is rotated further clockwise beyond the configuration of FIG. **26**, the drive wheel **642** reaches a centered position or centered configuration. The centered configuration may represent a fully retracted configuration, in which the control stops **330** have reached their most inward position. In the centered configuration, the axes **650**, **662**, **664**, **672**, and **674** may all be intersected by one line (e.g., the axis **620**). Accordingly, the forces F both act along a line intersecting the axis **650**, and the resultant moment M from the forces F is negligible.

[0097] In FIG. **27**, the drive wheel **642** is rotated further clockwise beyond the centered configuration. Accordingly, FIG. **27** represents an over center configuration of the control linkage **640**. The over center configuration may also be considered a travel configuration, in which the mower deck **80** is held at the travel height. In the over center configuration, the control stops **330**

may be extended slightly outward relative to the centered configuration. As shown, the shaft stops **322** each engage one of the control stops **330**, limiting outward movement of the actuator shafts **320**. The links **660** and **670** continue to apply a retracting force on the control stops **330**, placing the links **660** and **670** in tension. The links **660** impart a force F on the drive wheel **642**, and the links **670** impart a force F on the drive wheel **642**. However, in the over center configuration, the forces F are applied on an opposite side of the axis **650** relative to the configurations of FIGS. **25** and **26**. Accordingly, the moment M resultant from the forces F acts in a negative direction (i.e., clockwise) on the drive wheel **642**. However, the links **660** and **670** contact one another and/or the support shaft **646** in the over center configuration, acting as a hard stop and preventing further rotation in the clockwise direction. An extending force on the actuator shafts **320** (e.g., from the weight of the mower deck **80**) further loads against the hard stop, holding the control linkage **640** in the over center configuration. Accordingly, the control linkage **640** can remain in the over center configuration without application of a torque by the electric motor **440**. By utilizing the over center configuration, the mower deck **80** can be held in the elevated travel position mechanically, without continuous application of electrical energy.

[0098] To lower the mower deck **80**, the process shown in FIGS. **24-27** may be followed in reverse. The electric motor **440** may apply a torque to move the drive wheel **642** toward the center position. Once past the center position, the moment M acts in a positive direction. Accordingly, the mower deck **80** may be lowered by deactivating the electric motor **440** and permitting gravity to draw the mower deck **80** downward and extend the deck actuator **300**.

[0099] As utilized herein with respect to numerical ranges, the terms “approximately,” “about,” “substantially,” and similar terms generally mean $\pm 10\%$ of the disclosed values, unless specified otherwise. As utilized herein with respect to structural features (e.g., to describe shape, size, orientation, direction, relative position, etc.), the terms “approximately,” “about,” “substantially,” and similar terms are meant to cover minor variations in structure that may result from, for example, the manufacturing or assembly process and are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0100] It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0101] The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0102] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that

such variations are intended to be encompassed by the present disclosure.

[0103] The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules, and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single-or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

[0104] The present disclosure contemplates methods, systems, and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0105] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

[0106] It is important to note that the construction and arrangement of the mower **10** and the systems and components thereof (e.g., the body **20**, the operator controls **40**, the driveline **50**, the suspension system **60**, the braking system **70**, the control system **100**, etc.) as shown in the various

exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. By way of example, the transmission **500** may be utilized with the deck actuator **300** of FIG. 7 in place of the transmission **450**.

Claims

1. A mower, comprising: a chassis; a tractive element coupled to the chassis; a mower deck coupled to the chassis and including a cutting element; and a deck actuator assembly configured to raise the mower deck relative to the chassis, the deck actuator assembly including: a frame coupled to the chassis; an output interface coupled to the mower deck and slidably coupled to the frame; a first stop fixedly coupled to the output interface; a second stop positioned to (a) engage the first stop to limit movement of the output interface in a first direction and (b) permit movement of the first stop in a second direction opposite the first direction; and an actuator configured to reposition the second stop relative to the frame.
2. The mower of claim 1, wherein movement of the output interface in the first direction lowers the mower deck, and wherein movement of the output interface in the second direction raises the mower deck.
3. The mower of claim 1, wherein the actuator includes a threaded rod engaging the second stop and a motor configured to drive rotation of the threaded rod to reposition the second stop relative to the frame.
4. The mower of claim 3, wherein the threaded rod extends through an aperture defined by the first stop.
5. The mower of claim 3, further comprising a gear train coupling the motor to the threaded rod, the gear train including a first gear coupled to the motor and a second gear coupled to the threaded rod.
6. The mower of claim 5, further comprising a third gear coupling the first gear to the second gear.
7. The mower of claim 6, wherein the third gear is in meshing engagement with the first gear, and wherein a diameter of the third gear is larger than a diameter of the first gear.
8. The mower of claim 7, wherein the first gear is longitudinally offset from the second gear.
9. The mower of claim 6, wherein the threaded rod is a first threaded rod, wherein the actuator further includes a second threaded rod engaging the second stop, and wherein the third gear is coupled to the second threaded rod.
10. The mower of claim 5, wherein the first gear is in meshing engagement with the second gear.
11. The mower of claim 10, wherein the threaded rod is a first threaded rod, wherein the actuator further includes a second threaded rod engaging the second stop and a third gear coupled to the second threaded rod, and wherein the third gear is in meshing engagement with the first gear.
12. The mower of claim 1, further comprising an actuator shaft coupling the first stop to the output interface, wherein the actuator shaft extends through an aperture defined by the second stop.
13. The mower of claim 12, wherein the aperture is a first aperture, further comprising a guide defining a second aperture and a standoff extending between the guide and the frame, wherein the actuator shaft extends through the second aperture.
14. The mower of claim 13, wherein the standoff extends through a third aperture defined by the first stop and through a fourth aperture defined by the second stop.
15. The mower of claim 1, further comprising: a sensor coupled to the frame and configured to provide sensor data indicating a position of the first stop; and a controller operatively coupled to the actuator and the sensor and configured to limit operation of the actuator based on the sensor data.
16. The mower of claim 1, further comprising: a sensor coupled to the frame and configured to provide sensor data indicating a position of the second stop; and a controller operatively coupled to

the actuator and the sensor and configured to limit operation of the actuator based on the sensor data.

17. A mower, comprising: a chassis; a mower deck coupled to the chassis and including a cutting element; and a deck actuator assembly configured to raise the mower deck relative to the chassis, the deck actuator assembly including: a frame coupled to the chassis; an actuator shaft coupled to the mower deck and slidably coupled to the frame; a shaft stop fixedly coupled to the actuator shaft; a control stop positioned to engage the shaft stop, wherein the actuator shaft is configured to move freely relative to the frame throughout a longitudinal range of motion, and wherein an end of the longitudinal range of motion is defined by the shaft stop engaging the control stop; and an actuator configured to vary the longitudinal range of motion of the actuator shaft by repositioning the control stop relative to the frame.

18. The mower of claim 17, wherein the actuator includes: a threaded rod rotatably coupled to the frame and extending through a threaded passage defined by the control stop; and an electric motor coupled to the frame and configured to drive rotation of the threaded rod.

19. A floating linear actuator comprising: a frame defining a first connection interface; a guide defining a shaft passage, the shaft passage extending longitudinally through the guide; a standoff extending longitudinally between the frame and the guide and fixedly coupling the guide to the frame; an actuator shaft defining a second connection interface and extending through the shaft passage; a shaft stop fixedly coupled to the actuator shaft; a control stop defining a threaded passage, the control stop being positioned to engage the shaft stop to limit movement of the shaft stop in a first direction and permit movement of the shaft stop in a second direction; a threaded rod rotatably coupled to the frame and extending through the threaded passage; and a motor configured to drive rotation of the threaded rod.

20. The floating linear actuator of claim 19, wherein the first direction is an extending direction, wherein the second direction is a retracting direction, and wherein the movement of the actuator shaft in the extending direction increases a distance between the first connection interface and the second connection interface.
